PROJECT PROPOSAL

IMPLEMENTING MATHEMATICAL MODELS FOR EARTHQUAKE EPICENTER ESTIMATION: A DATA - DRIVEN SOFTWARE SYSTEM FOR HISTORICAL AND CONTEMPORARY SEISMIC EVENTS

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DATE OF SUBMISSION:

SUBMITTED TO:

KEYWORDS:

PROJECT SUMMARY

Historical earthquakes, which occurred before modern seismometers, remain crucial for understanding long-term seismic risk. Estimating their epicenters, however, is challenging since only descriptive intensity data are available. Established approaches—such as generalized isoseismal maps, tectonic structures, and analogies to recent seismicity—have provided valuable insights for reconstructing historical events. This project seeks to complement these methods by introducing a mathematical model developed at the Mathematics Department of Ateneo de Davao University that estimates epicenters directly from intensity observations, including those derived from historical accounts. The approach fits circles to locations reporting the same intensity, applies statistical optimization to determine the most probable epicenter, and quantifies uncertainty through confidence intervals.

The project unfolds in three stages: (1) validating the model on recent, well-documented earthquakes; (2) applying it to historical earthquake data to enrich seismic catalogs; and (3) developing an interactive software platform integrating these methods with earthquake databases, hazard maps, and optional AI/ML tools for pattern recognition.

This initiative strengthens existing seismological practices, providing researchers and policymakers with additional tools for hazard assessment, educational use, and disaster preparedness.

I. BACKGROUND AND RATIONALE

Earthquake hazard assessment is a critical priority for disaster risk reduction in the Philippines, one of the world's most seismically active regions. Historical earthquakes, many of which occurred before the establishment of modern seismic networks, remain essential for understanding long-term seismicity patterns. However, the absence of instrumental data poses a challenge for accurately determining their epicenters and integrating them into hazard maps.

Traditional approaches — such as isoseismal map interpretation, tectonic correlation, and analog comparison with recent events — have been invaluable, yet they often rely on expert judgment and qualitative assessments. There is a need for complementary, quantitative methods that provide reproducible and statistically robust results.

This project addresses that gap by developing and validating a mathematical model that estimates earthquake epicenters directly from intensity observations, which can be reconstructed from historical accounts. The model applies circle-fitting and statistical optimization techniques to produce epicenter estimates with quantified uncertainty. The project will proceed in three stages: (1) validation of the mathematical method using modern earthquakes with known epicenters, (2) application to historical earthquake records to refine seismic catalogs, and (3) development of a software platform integrating these methods with earthquake catalogs, hazard maps, and potential AI/ML tools for enhanced pattern recognition.

This initiative offers an innovative, data-driven approach to improving earthquake catalogs and hazard models by combining applied mathematics, computer science, and seismology. Its outcomes are expected to support PHIVOLCS, LGUs, and researchers in strengthening disaster preparedness, risk communication, and resilience planning at both local and national levels.

II. OBJECTIVES

Objectives of the Project:

This project seeks to create a robust, data-driven software platform that unifies mathematical modelling, seismological data, and computational tools to improve earthquake epicenter estimation. Its ultimate goal is to enhance the scientific understanding of seismic events while providing accessible, actionable tools for researchers, disaster risk managers, and educators.

General Objective:

To design and deploy an integrated software system that applies mathematically rigorous models for earthquake epicenter estimation, enabling accurate, reproducible, and user-friendly processing and visualization of historical and contemporary seismic data.

Specific Objective/s:

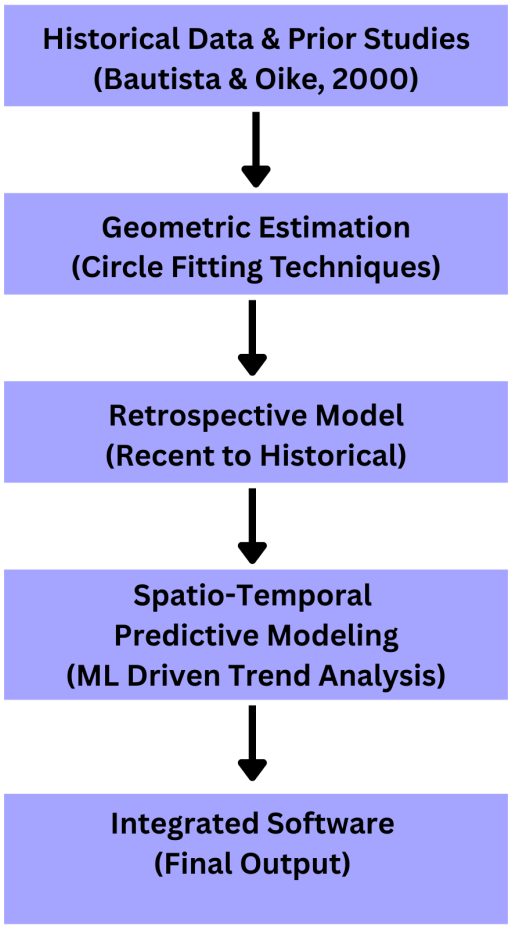
To achieve this, the project aims to:

1. Formulate and implement mathematical models
2. Evaluate the model's reliability and precision
3. Develop a software platform
4. Provide interactive geospatial tools
5. Establish an internal validation
6. Design a secure and scalable data management system

III. FRAMEWORKS AND METHODOLOGY

Conceptual Framework

This project aims to develop a software system that estimates earthquake epicenters based on intensity or magnitude observations, particularly for historical earthquakes with no instrumental data. The framework integrates four disciplines - mathematics (theories and methods such as circle fitting and modelling), seismology (interpretation of intensity data and validation), data science (pattern recognition and backward forecasting), and computer science (software development).



Flow of the Project

Input: Historical intensity reports, recent seismic events, and catalogs (PHIVOLCS, Ateneo Observatory)

Processing: **(1)** Mathematical modelling - circle fitting, least-squares estimation, and uncertainty intervals, **(2)** Machine Learning - backward modeling of historical seismicity, clustering of epicenters, forecasting future seismic trends, **(3)** Seismological validation - Interpretation and cross-checking records and plausibility.

Output: Most probable epicenter coordinates, Confidence regions, Interactive maps and data visualization, Reports for research and hazard mapping, Probabilistic analysis of future trends

Core Ideas

The following are the core ideas present in the project:

1. Bautista and Oike (2000) - "Estimation of the magnitudes and epicenters of Philippine historical earthquakes"

   This work established methods for estimating magnitudes and epicenters of historical Philippine earthquakes. Their approach combined felt-area - magnitude relationships, isoseismal maps, and tectonic information to reconstruct past events. We adopt these principles as the foundation of our historical catalog, ensuring that our reconstructions align with national seismological standards and provide a reliable dataset for subsequent modeling.

2. Umbach and Jones (2004) - Circle Fitting Methods

   These methods allow us to convert intensity observations into reproducible intensity observations into reproducible geometric models by fitting circles to locations reporting the same intensity level. The center of each circle represents the estimated epicenter, while the radius represents the extent of the shaking. We generate epicentral estimates with quantifiable uncertainty by applying iterative refinement and weighting based on data reliability - an essential input for our Bayesian error-refitting process.

3. Retrospective modeling

   The project will work backwards from well-instrumented recent events toward older, less specific historical earthquakes. By calibrating our models using modern seismic data, we iteratively refine earlier estimates, producing a coherent spatiotemporal sequence of earthquake epicenters. This backward-looking approach ensures that the historical record benefits from the improved resolution of recent data.

4. Spatiotemporal Predictive Modeling

   Transform the reconstructed earthquake catalog into a machine learning and statistical modeling data set. Features such as location, magnitude, time intervals, and clustering patterns are used to identify spatiotemporal trends and generate probabilistic

forecasts of likely epicentral zones. This step complements the existing hazard mapping efforts by providing a data-driven, adaptive layer of analysis that can evolve as more data becomes available.

Methodology

Phase 1: Data Collection and Preprocessing

- **Sources:** Historical intensity reports (archival records), PHIVOLCS bulletins, peer-reviewed catalogs.
- **Cleanings:** Standardize coordinates, assign numeric intensity scales, filter inconsistent entries.
- **Preprocessing:** Construct intensity distribution maps and group data by event and prepare them for modeling.

Requirements Gathering and Analysis

The initial phase of the study will involve gathering and analyzing comprehensive requirements. System requirements will be elicited in consultation with domain experts, including seismologists from PHIVOLCS and mathematicians from the Mathematics Department. Particular attention will be given to understanding user needs regarding data input, epicenter estimation, historical catalog management, and interactive visualization. Institutional constraints on data storage and confidentiality, including policies concerning on-premise deployment or potential use of cloud services, will also be considered. These requirements will guide the design specifications and inform system architecture and module functionalities decisions.

Phase 2: Mathematical Modeling

- **Techniques:** Bautista & Oike (2000), Umbach & Jones (2004) Circle - Fitting Algorithms. Optimization to minimize distance error between reported intensities and fitted circles.
- **Outputs:** Estimated epicenter coordinates, Confidence radii, and intervals to quantify uncertainty.

The proposed methodology begins by fitting the reported intensity data to a circle-fitting model, allowing the most probable epicenter coordinates to be computed. Once the initial estimate is obtained, a Bayesian framework will be applied to refine the result by incorporating uncertainties and adjusting the model parameters based on the observed error distribution. This iterative process produces a statistically robust estimate of the earthquake epicenter with a quantified confidence.

Phase 3: Machine Learning and Data Analysis

Goals:

- Backward modeling: Reconstruct past patterns using intensity-based estimates.
- Detect spatio-temporal clusters of historical events.
- Apply regression or probabilistic models to infer trends in earthquake epicenters over time.

Techniques:

- Retrospective Modeling
- Spatio-temporal Modeling

Following Phase 3, a retrospective modeling approach will be implemented, beginning with well-documented recent earthquakes and progressively working backward to historical events. This approach enables validation of the model against known data and calibration for older records. Finally, the project will leverage the retrospective model outputs to perform spatio-temporal analyses, identify seismic patterns, and generate insights into potential future

trends in earthquake occurrence, offering a valuable tool for hazard assessment and disaster preparedness.

Retrospective Modeling (Candidate Models)

1. Bayesian Inference with Hierarchical Modeling

   Bayesian Inference with Hierarchical Modeling will integrate multiple sources of uncertainty, including historical intensity reports, catalog completeness, and location accuracy. This will be applied during data analysis and parameter estimation. This is done so that.

   A. Jointly estimate earthquake magnitude and location distributions while accounting for uncertainty in historical data.
   B. Allow probabilistic inference on unobserved parameters (e.g., epicentral intensity) using prior information from existing catalogs.

   Improves reliability of parameter estimates by providing confidence intervals, making outputs more robust for use in hazard assessments.

2. Inverse Problem Approach with Circle-Fitting with Maximum Likelihood

   Adopt and improve circle fitting techniques and combine with maximum likelihood estimation to reconstruct possible epicenters from observed isoseismal radii. This will be applied in isoseismal data reconstruction and epicentral estimation.

   A. Fit circles to intensity data points to approximate isoseismal contours on maps.
   B. Use likelihood-based optimization to refine epicentral location and magnitude estimates.

   Provides a procedure for locating historical earthquakes using sparse or unevenly distributed intensity data.

3. Space-Time Point Process Models (ETAS/Hawkes Process)

   Utilize Epidemic-Type Aftershock Sequence (ETAS) models or Hawkes processes to analyze the temporal clustering of historical earthquakes. This will be applied in spatiotemporal modeling and predictive analysis.

   A. Speculate on the probable future events given past events.
   B. Identify spatiotemporal clustering patterns (e.g., epicentral zones, migration of seismicity).

   Helps uncover hidden patterns in historical sequences and generate probability surfaces for future event occurrences.

Spatio-Temporal Modeling (Candidate Models)

1. Space-Time ETAS (Epidemic-Type Aftershock Sequence) Models

   The space-time ETAS model is a parametric point process model that characterizes earthquake triggering dynamics.

   A. Model how historical mainshocks triggered aftershocks in space and time.
   B. Infer potential epicentral zones by tracing back clusters of activity.

2. Spatio-Temporal Point Process Models (Non-Parametric / Kernel Based)

   Kernel-based point process models to flexibly estimate earthquake intensity rates without assuming a strict parametric form.

   A. Construct smooth intensity surfaces over space and time.
   B. Identify historical “hotspots” of seismicity where records may be incomplete.

   Allows for exploratory analysis and visualizations of historical seismic data before committing to parametric models, supporting hypothesis generation.

3. Bayesian Hierarchical Spatio-Temporal Model

This model unifies spatial and temporal dependencies under a Bayesian framework. This applies in joint parameter estimation and uncertainty quantification.

A. Estimate time - varying seismicity rates with spatial correlations.
B. Propagate uncertainty across all layers (data to parameters to predictions). Produces probabilistic forecasts and confidence intervals.

4. Gaussian Process Spatio-Temporal Regression

Gaussian Process Regression (GPR) will model intensity fields as continuous spatio-temporal functions. This will be applied in isoseismal surface reconstruction and prediction.

A. Interpolate intensity values in space-time where observations are missing.
B. Predict intensity at unobserved locations/times for validation of reconstructed historical events.

Enables spatio-temporal mapping, turning historical observations into smooth surfaces suitable for further modeling.

Phase 4: Software Development

System Design

The system design will adopt a modular architecture to ensure separation of concerns and maintainability. Core modules include:

1. Epicenter Estimation Module: Responsible for computing probable epicenters and quantifying associated uncertainties.
2. Historical Earthquake Catalog Integration: Handles the organization of historical earthquake catalogs and manages the input and storage of intensity reports.

3. Geovisualization Layer: Provides interactive, map-based visualizations of computed epicenters and associated data.
4. Comparative Validation Tool: Facilitates cross-referencing of computed epicenters with official records from PHIVOLCS.
5. Machine Learning / AI Integration: Responsible for anomaly detection in spatio-temporal seismicity patterns.

To support computationally intensive tasks, a Python microservice is integrated for performing epicenter computations, including handling large datasets containing hundreds to thousands of data points per calculation. The NestJS backend communicates with the Python microservice via RESTful APIs, transmitting input data and receiving computed epicenter coordinates, proximity radii, and uncertainty measures. Raw SQL queries are employed for database interactions to optimize performance for large datasets, and incremental update mechanisms ensure that newly added data triggers updates only for affected computations.

System Development

The system will be implemented as a web-based platform. The frontend, developed using Next.js, will provide a responsive and intuitive interface for researchers to input intensity reports, view computed epicenters, and interact with historical catalogs. The backend, implemented in NestJS, will be the main API layer, coordinating computations with the Python microservice and managing database interactions. The Python microservice will execute core computational routines for epicenter estimation, statistical analyses, and optional anomaly detection features. Incremental computation will be supported, allowing efficient updates when new intensity reports are added without recomputing the entire dataset.

Testing and Validation

Unit testing and integration testing will be conducted on all system modules to ensure correctness and reliability. Accuracy testing will involve applying the system to recent Philippine earthquakes with known epicenters and comparing computed results against official PHIVOLCS

records. Usability testing will be performed to obtain feedback from potential users, ensuring the interface is user-friendly and accessible.

Documentation and Deployment

The final phase will focus on documentation and deployment. Technical documentation, including user manuals and developer references, will be prepared to facilitate effective system use and future maintenance. Deployment strategies will align with institutional policies, allowing for on-premise server installation or potential cloud-based hosting. Training and knowledge transfer sessions will be conducted to equip researchers and institutional staff with the necessary skills to operate the system effectively.

Proposed System Features

The proposed system will be a web-based platform to facilitate earthquake epicenter estimation, historical catalog management, and interactive geovisualization. The architecture will integrate a Next.js frontend, NestJS backend, and a Python microservice for computationally intensive tasks. All backend and microservice communications will be handled via REST APIs using JSON.

Epicenter Estimation Module

The Epicenter Estimation Module will serve as the core computational component of the system. It will accept earthquake intensity reports, including latitude, longitude, intensity, and associated metadata. Computation will be delegated to the Python microservice, processing thousands of data points per run to calculate epicenter coordinates and corresponding uncertainty measures. The system will support incremental updates, allowing newly added data to trigger partial recomputation rather than processing the entire dataset. Additionally, chunked computation will be employed for nationwide visualizations to reduce computational load. The module will return JSON payloads containing the estimated epicenter coordinates, proximity

radius, confidence intervals, and contributing observation identifiers, which other modules will then use for display and validation.

Historical Earthquake Catalog Integration

The Historical Earthquake Catalog Integration module will store and organize records obtained from the Ateneo Observatory Archives and PHIVOLCS. The preprocessing pipeline focuses on cleaning the dataset by handling missing values, removing duplicates, and standardizing data-related features to ensure consistency and reliability for analysis. This module will provide a searchable interface for researchers, enabling filtering and timeline navigation across historical events.

Geovisualization Layer with LeafletJS

For geovisualization, the system will employ LeafletJS to allow users to explore earthquake epicenters on map-based outputs interactively. Epicenters will be represented as markers color-coded by confidence level, with optional popups displaying event details, contributing observations, and uncertainty radii. Circles will be drawn around epicenters to illustrate proximity or uncertainty, while MarkerCluster will handle dense historical datasets to reduce visual clutter. The map layer will support interactive filtering by date, intensity, region, and confidence level. Incremental updates will ensure that newly computed epicenters or data points can be dynamically added without re-rendering the entire map.

Comparative Validation Tool

The Comparative Validation Tool will enable computed epicenters to be cross-referenced with official PHIVOLCS records. Researchers may also download validation reports for further analysis.

Machine Learning / AI Integration

The integration of Machine Learning and Artificial Intelligence enables anomaly detection by identifying irregularities and uncovering emerging patterns within historical and current datasets. A preprocessing pipeline ensures data quality through systematic cleaning, standardization, and transformation before model training. The system then applies methods such as Autoencoders, clustering algorithms (k-means, DBSCAN), and anomaly detection techniques (Isolation Forest, LOF) to detect deviations and spatio-temporal patterns using inputs like magnitude, intensity, location, depth, and time, supported by auxiliary data such as soil type and population density.

System Architecture Overview

The proposed system will follow a modular, microservice-oriented architecture that separates computationally intensive tasks from the main web application while maintaining seamless integration. The frontend will be implemented in Next.js, serving as the user interface and providing panels for data input, historical catalog access, epicenter visualization, and validation overlays. User actions on the frontend will trigger requests to the backend, implemented in NestJS, which will manage authentication, request routing, data retrieval, and integration with the database.

The Python microservice will execute the mathematical algorithms for epicenter estimation, including circle-fitting, statistical analysis, and processing thousands of observational points per computation. Communication between the NestJS backend and the Python microservice will be facilitated through RESTful APIs using JSON payloads. This design will allow incremental and chunked computations, reducing computational overhead when processing nationwide datasets or updating existing epicenters with newly added observations.

The database will be the central repository for historical earthquake catalogs, computed epicenter results, and associated metadata. Raw SQL queries will be employed for optimized performance when retrieving large datasets, while the backend will ensure data integrity, standardization, and security per institutional policies.

A high-level architecture diagram will illustrate the interactions between the frontend, backend, Python microservice, and database. The architecture will ensure scalability, maintainability, and ease of integration for future enhancements, such as advanced machine learning modules for anomaly detection.

Phase 5: Validation and Testing

- **Comparison:** Cross-validate computed epicenters with official PHIVOLCS locations and published studies.
- **Expert Review:** Invite seismologists and external reviewers to critique results and refine methodology.

The developed software will be tested against real-world data, feedback will be gathered on how useful the software is to the seismologist, and user experience will be evaluated for seismologists and related researchers. Afterwards, a consultation with an expert will be conducted for further insights, plausibility, feedback, recommendations, and review.

Phase 6: Dissemination and Impact

- **Outputs:** Conference papers, journal submissions, and project reports.
- **Stakeholders:** PHIVOLCS, local disaster risk reduction offices, and academic institutions.
- **Applications:** Enhanced hazard maps, improved preparedness plans, and historical earthquake reinterpretation.

After the software is turned over and the project is completed, the project can be presented in conference proceedings and finalized for publication.

IV. EXPECTED OUTPUTS

1. **Validation of Circle-fitting Methodology and Reconstruction of Epicenter/Magnitude Estimates for Selected Historical Earthquakes in the Philippine Seismic Context.** The circle-fitting methodology will be tested on recently

recorded earthquakes to achieve validation. The resulting computation of the epicentral coordinates will be validated against known epicenters to determine the model's accuracy. Once its accuracy is validated, the circle-fitting method will be executed on historical earthquakes in the Philippines, generating new epicenter estimates for each earthquake.

2. **Development of a Fully Functional Prototype Software Platform Integrating Mathematical Models and Data Visualization.** This project operationalizes the mathematical model with a functional software platform. This software primarily generates epicentral coordinates with quantified uncertainty and an interactive map-based visualization for analytical use.
3. **Scholarly Contributions and Student Mentorship Outcomes.** An academic publication of this system would lead to a permanent record of the proponents' work. The interdisciplinary research and application of mathematical models and software engineering contribute to the scientific community. Moreover, students would benefit from mathematical modeling, data science, and multidisciplinary teamwork training by studying mathematical modeling and software engineering.

V. IMPACTS AND APPLICATIONS

Accurately estimating earthquake epicenters is vital to seismology, as it underpins hazard assessment, disaster preparedness, and risk reduction strategies. While modern methods depend heavily on instrumental data, many earthquakes of historical importance occurred before the availability of such technology. This leaves a significant gap in the seismic record, particularly in regions like the Philippines, where historical earthquakes have shaped the physical landscape and the resilience of communities. This study is significant for several reasons:

- Contribution to Seismology. The study provides an alternative method for interpreting earthquakes that lack instrumental data by operationalizing a mathematical model that derives epicenter estimates from intensity observations. This enhances the capacity to reassess historical earthquakes with greater accuracy.

- Practical Application for Institutions. The development of a software tool offers PHIVOLCS, the Ateneo Observatory, and other research institutions a valuable platform for cataloging, analyzing, and visualizing earthquake data. The tool complements existing methods, serving as a cross-validation mechanism for official epicenter determinations.
- Interdisciplinary Value. This project demonstrates the benefits of collaboration between mathematics, computer science, data science, and seismology. By combining algorithmic modeling with software design, the study creates a bridge between theoretical methods and practical applications.
- Global Relevance. Although the study emphasizes the Philippine context, the methodology and software design can be adapted to other regions with available intensity data. This gives the project a broader relevance to international seismological research and historical earthquake studies.

VI. TEAM COMPOSITION

The interdisciplinary project will cover mathematics, seismology, computer science, and data science. It will comprise professionals from the Ateneo de Davao Mathematics Department and an expert from the Philippine Institute of Volcanology and Seismology (PHIVOLCS). Meanwhile, it will be composed of students from Ateneo de Davao University currently in undergraduate studies in Computer Science, Data Science, and Computer Engineering degrees. This is presented below in alphabetical order.

Mathematics Department

Buelban, Orville B.
Jamesolamin, Ronaldson Jr.
Tura, Ric Jr

Role: Theoretical Modelers and Solid, Rigorous Backbone

Contribution/s:

A. Develop and refine algorithms or methods to estimate historical and recent epicenters.
B. Provide mathematical rigor, collaborate with Data Science Students to ensure proper statistical assumptions and model validity.

C. Support derivation of confidence intervals, error bounds, and theoretical consistency across methods.

DOST - PHIVOLCS

Perez, Jeffrey

Role: Domain Experts and Validation

Contribution/s:

A. Suggest or supply historical earthquake catalogs and recent seismic event data for model calibration and validation.
B. Provide expertise in seismology or geology, ensuring models reflect real-world situations.
C. Validate reconstructed epicenters and intensity maps.
D. Act as advisors on practical impact, ensuring project outputs align with national hazard assessment goals.

Data Science

Mamac, Ira Zaky – 3rd Year BS Data Science, Ateneo de Davao University

Sulit, Zachary Lorenzo – 3rd Year BS Data Science, Ateneo de Davao University

Role: Assistant Modelers and Data Analyst

Contribution/s:

A. Assist and implement retrospective modeling and spatio-temporal modeling efforts.
B. Perform statistical validation, cross-validation.
C. Implement machine learning components for spatio-temporal trend detection and forward-looking predictions.

Computer Science

Aventura, Arielle Eliza Denise – 3rd Year BS Computer Science, Ateneo de Davao University

Calong, Kirby Josh – 3rd Year BS Computer Science, Ateneo de Davao University

Liu, EJ – 3rd Year BS Computer Science, Ateneo de Davao University
Obeso, Chastin Karol – 3rd Year BS Computer Science, Ateneo de Davao University
Regalado, Derren Hanz – 3rd Year BS Computer Science, Ateneo de Davao University
Tan, Kyle Daniel – 3rd Year BS Computer Science, Ateneo de Davao University
Tesoro, Sam Adrianne – 3rd Year BS Computer Science, Ateneo de Davao University

Role: Implementation of methods via software development

Contribution/s:

A. Build the frontend and backend infrastructure for the earthquake modeling and visualization platform.
B. Create interactive dashboards and visualizations for epicenter estimation and results, isoseismal reconstructions and hazard maps.
C. Ensure scalability and maintainability of the platform.

Computer Engineering

Teves, Fergievon Alaika – 3rd Year BS Computer Engineering, Ateneo de Davao University

Role: Implementation of methods via software development

Contribution/s:

D. Build the frontend and backend infrastructure for the earthquake modeling and visualization platform.
E. Create interactive dashboards and visualizations for epicenter estimation and results, isoseismal reconstructions and hazard maps.
F. Ensure scalability and maintainability of the platform.

VII. RESOURCE REQUIREMENTS

The successful execution of this project requires a combination of hardware, software, human expertise, data access, and institutional support.

1. **Hardware Resources**
    - Development server or cloud-based virtual machine with at least 8 GB RAM, a multi-core processor, and 100 GB storage for handling earthquake datasets.
    - Standard laptops/desktops (≥8 GB RAM) for researchers to conduct coding, analysis, and offline testing.
    - Stable internet connectivity to support deployment, real-time collaboration, and software updates.
    - Networking equipment and backup drives for data security and redundancy.

2. **Software Resources**
    - Programming and Data Analysis: Python (NumPy, SciPy, Pandas) for model implementation and validation.
    - Web Development: Next.js for the frontend; Supabase for backend and database integration.
    - Deployment and Testing: Docker for containerized deployment, Postman for API testing, GitHub for version control and collaboration.
    - UI/UX Design: Figma for interface prototyping and usability design.
    - Documentation Tools: Overleaf/LaTeX and Google Docs for technical writing, reporting, and academic publications.

3. **Human Resources**
    - *Lead Researcher / Project Leader* – oversees methodology, ensures alignment with objectives, and coordinates the team.
    - *Developers* (Frontend, Backend, Full-stack) – implement system modules and maintain the web platform.
    - *Data Analyst / Data Scientist* – performs statistical validation, model testing, and optimization.
    - *UI/UX Designer* – ensures system usability and accessibility.
    - *Domain Experts / Consultants* – collaboration with seismologists (e.g., PHIVOLCS) and academic mentors for technical validation.

- *Research Assistants* – support data preprocessing, catalog digitization, testing, and documentation.

4. **Data Resources**
    - Historical earthquake catalogs from Ateneo Observatory Archives and PHIVOLCS.
    - Recent seismic event data for model validation and benchmarking.
    - Metadata from intensity reports (location coordinates, magnitude, descriptions).
5. **Institutional and Logistical Resources**
    - Research workspace equipped with computers, stable internet, and meeting facilities.
    - Access to university library resources, journals, and prior earthquake studies.
    - Institutional agreements or Memoranda of Understanding (MoUs) with PHIVOLCS and Ateneo Observatory for data access.
    - Budget for travel, workshops, and conferences to support collaboration and dissemination of results.
6. **Contingency Resources**
    - Additional cloud compute credits for large-scale dataset processing.
    - Additional research assistants in case of extended digitization or testing tasks.
    - Reserve budget for proprietary software tools, emergency hardware repair, or unplanned expenses.

PROJECTED BUDGET

Category	Description	Estimated Cost (PHP)
Personnel Costs	Researcher and consultant's allowance (seismology & mathematics), technical support.	300,000 – 500,000
Software Requirements	Open-source frameworks (Next.js, NestJS, Python, LeafletJS, SQL); minimal	0 – 30,000

	optional tools	
Hardware Requirements	Application + database servers, networking, redundancy	280,000 – 450,000
Hosting (Cloud option)	Scalable storage and computation (alternative to on-premise)	10,000 – 25,000/month
Database Considerations	Storage, optimization, backup systems	Included in hardware/hosting
Miscellaneous / Contingencies	Travel, workshops, peripherals, documentation, dissemination	50,000 – 100,000
Total Estimated Cost (Year 1)	Personnel + Hardware + Software + Miscellaneous (excluding recurring cloud hosting)	~630,000 – 1,080,000

Remark: Computer servers are required for the project, but it would be too costly to build, operate, and maintain them. To minimize cost as well as maintain the privacy and ownership of data, we would need existing computer servers (if there are any) from DOST. It is also impractical to use Ateneo de Davao University servers due to the amount of computational resources.

VIII. PROJECT TIMELINE

Activity	**Estimated Duration**	**Start Date**	**End Date**	**Deliverable**
Proposal	1 month	August 2025	September 2025	Written project proposal document
Validation of	1 month	September	Early	Validation report

circle-fitting on recent data		2025	October 2025	
Historical earthquake application	1 month	Mid October 2025	Late October 2025	Use historical earthquake data for model validation
Development kickoff	1 month	November 2025	November 2025	Project plan & team alignment report
Ateneo de Manila University conference	1 day	December 2025	December 2025	Conference presentation slides
Completion of working prototype	3 months	January 2025	March 2025	Functional prototype build
Quality assurance testing of the application	1 month	March 2025	April 2025	Extensive testing
Presentation of the prototype	1 day	April 2025	April 2025	Prototype demo & presentation
Continued development	4 month	May 2025	September 2025	Updated application version
Integration and testing with a real data set	1 month	September 2025	October 2025	Integration test results
Publication, dissemination, and reporting	2 month	October 2025	December 2025	Final report & published paper